

# Erdős Problem #706

|        |                |                      |              |
|--------|----------------|----------------------|--------------|
| EP-706 | L1 - Tractable | Partiellement résolu | Graph Theory |
|--------|----------------|----------------------|--------------|

## Énoncé original (anglais, reproduction mot pour mot)

Let  $L(r)$  be such that if  $G$  is a graph formed by taking a finite set of points  $P$  in  $\mathbb{R}^2$  and some set  $A \subset (0, \infty)$  of size  $r$ , where the vertex set is  $P$  and there is an edge between two points if and only if their distance is a member of  $A$ , then  $\chi(G) \leq L(r)$ .

Estimate  $L(r)$ . In particular, is it true that  $L(r) \leq r^{O(1)}$ ?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>